The Direct Answer Dilemma

How AI Search Closes the Attribution Loop — and Who Pays for It

Audit synthesis

Potthast et al. (2020), *SIGIR Forum* | Liu, Zhang & Liang (2023), *arXiv:2304.09848*
Jazwinska & Chandrasekar (2024, 2025), *Tow Center / CJR* | Aggarwal et al. (2024), *KDD '24*
Khosravi et al. (2026), *arXiv preprint* | Larsen et al. (2025) | Liang et al. (2025), *Patterns*

60%

of citation queries answered *wrong* by eight AI search engines.

Grok 3: 94% wrong. Most wrong answers came with no hedging.

The interface earned the user's trust. The citations did not.

Jazwinska & Chandrasekar (2025), *Tow Center / CJR*; 1,600 queries across 8 chatbots.

Potthast (2020) named four risks of the direct-answer interface — all four now have data

Potthast risk (2020)	Mechanism	Empirical confirmation
Misattribution	Answer drops the source	Liu 23; Jazwinska 24/25
Source erosion	Click never happens	Khosravi 2026
Market distortion	Ad auction reshapes	Larsen 2025
Loop pollution	Synthetic text re-enters web	Aggarwal 24; Liang 25

Potthast, Hagen & Stein (2020), *SIGIR Forum* 54(1):14.

Citation is broken at the source

Direct-answer interfaces grade their own homework.

The better the answer reads, the worse it cites — r = -0.96

Engine	Citation recall	Citation precision
Bing Chat	58.7%	89.5%
NeevaAI	67.6%	72.0%
perplexity.ai	68.7%	72.7%
YouChat	11.1%	63.6%
Mean (4 engines)	**51.5%**	**74.5%**

Across 1,450 statements: Pearson $r = -0.96$ between citation precision and the perceived fluency/utility of the answer. Better-sounding answers cite worse.

Liu, Zhang & Liang (2023), Tables 3–4; human eval over four generative search engines, May 2023.

ChatGPT Search misattributed 153 of 200 publisher quotes — and declined only 7 times

Response category	Count / 200	Share
Correct (publisher + date + URL)	47	23.5%
Partially correct	57	28.5%
Wrong	89	44.5%
Declined / unsure	7	3.5%

Even publishers *with* OpenAI licensing deals were misattributed. NYT (in active lawsuit, blocked) was cited via plagiarized copies.

Jazwinska & Chandrasekar (2024), *Tow Center / CJR*, Nov. 2024; 200 quotes, 20 publishers.

Across 8 chatbots, premium tiers are *more* confidently wrong

Engine	**Inaccuracy**	**Behavior**
Perplexity (free)	**37%**	best of eight
ChatGPT Search	67%	hedged 15/200
Perplexity Pro (paid)	higher	decline rate falls
Gemini	high	fabricated URLs frequent
DeepSeek	high	misattributed 115/200
Grok 3 (paid)	**94%**	154/200 broken URLs

Paid tiers answer *more* prompts confidently — so they make *more* confident errors. Robots.txt routinely violated.

Jazwinska & Chandrasekar (2025), *CJR*, Mar. 2025; 1,600 queries (20 publishers × 10 quotes × 8 chatbots).

The economic price falls on the source

Direct answers are paid for in someone else's traffic.

AI summaries cut Wikipedia pageviews ∼15%, costing $35.7M–$102.1M per year

Outcome	Estimate	Source
Pageview decline (post-AI summaries)	-15%	DiD, 2024–25
Annual donation revenue at risk	$35.7M–$102.1M	extrapolation
Editor-recruitment funnel	weaker	traffic-volunteer link

The first feedback loop: fewer readers → fewer donations and editors → a thinner Wikipedia → a thinner training corpus.

Khosravi et al. (2026); pageview DiD around AI Overviews / Search GPT rollouts.

Sponsored-search auctions splinter: short-query CPC $+1.5\%$, long-query CPC -2.5%

Query length	Δ CPC (BERT)	Mechanism
Short (1–2 tokens)	$+1.5\%$	better intent match $\rightarrow$ tighter rivalry
Long-tail (5+ tokens)	-2.5%	query absorbed by AI answer

The second feedback loop: the highest-intent traffic that publishers monetized via long-tail SEO is exactly what AI answers consume — so its auction price falls.

Larsen et al. (2025); 11,949 queries $\times$ 24 months SEMRush panel; BERT rollout DiD.

Creators adapt — and the well gets polluted

Optimization moves on. The optimization target is now an AI.

Adding a single quotation lifts AI-engine visibility by 41%

GEO tactic	Δ visibility	Cost to publisher
Quotation Addition	+41%	low (one edit)
Statistics Addition	+32%	low
Authoritative tone	+30%	stylistic only
Citing Sources	+27%	low

Generative Engine Optimization (GEO) is now a discipline. The new ranking signal rewards *quotability*, not authority — so creators feed the engine what it likes.

Aggarwal et al. (2024), *KDD '24*; benchmark of 9 GEO tactics on 10K queries.

By late 2024, $\sim$24% of corporate press releases and $\sim$18% of CFPB complaints are LLM-written

Corpus	LLM share	Sample (N)
Corporate press releases	$\sim$24%	537,413
Consumer complaints (CFPB)	$\sim$18%	687,241
UN press releases	$\sim$14%	15,919
Small-firm job postings	$\sim$10–15%	1.44M

Distributional MLE; prediction error $< 3.3\%$; *lower bound* (heavy edits invisible). The corpus on which the next models train is increasingly authored *by* models.

Liang et al. (2025), *Patterns*; four corpora, Jan 2019–Sep 2024.

Devil's advocate: aren't most queries the kind where citations don't matter?

The strongest objection

For "what time is the show," "capital of Peru," "recipe for risotto," AI answers users more cheaply than ten blue links — and a wrong citation costs nothing.

Reply. Three problems with the carve-out:

- Liu (2023): $r = -0.96$ between fluency and citation precision — users have *no signal* that flips them from "trust" to "verify."
- Jazwinska (2025): premium-tier engines decline less, so confident wrongness rises with willingness to pay.
- Khosravi (2026) + Larsen (2025): even "trivial" queries strip the long-tail traffic that funds the verifiable corpus.

The cheap query subsidises the expensive one. Sever the cross-subsidy, both fail.

Potthast's prophecy is now a self-reinforcing loop with five turns of the crank

#	Turn	Empirical anchor
1	Citations decay	Liu 2023: precision 74.5%, recall 51.5%, $r = -0.96$
2	Misattribution scales	Jazwinska 24/25: 60%–94% wrong (8 engines)
3	Page views collapse	Khosravi 2026: -15% Wikipedia traffic
4	Ad auctions reprice	Larsen 2025: -2.5% long-tail CPC
5	Synthetic content fills gap	Aggarwal 24 (GEO $+41\%$); Liang 25 ($\sim$24% PR)

The training corpus that produced the model is being eroded by the model. Each turn cheapens the next.

Direct answers are not free.

They are an externality on the web that produced them.